Applicant Ranking (Lever ATS + LLM)

Egen internal project for Talent Acquisition, started 2026-10-08. The requirements are in docs/requirements.pdf.

- **Main contact:** Mike Brinn
- **Users:** Egen Talent Acquisition (recruiters)

Status: milestone 1 built (2026-10-08)

Core library, CLI and webhook service, with about 2,500 tests run against mocks (2,518 on 2026-10-12). It has not yet run against the real Lever or Gemini APIs: that needs an Egen GCP project and a Lever API key (see Before first real run).

- docs/requirements.pdf: what it must do
- docs/design.md: how it works and why
- docs/legal-hr-questions.md: decisions needed before live use
- docs/research-notes.md: verified facts and sources

Quick start

```
uv sync
```

```
uv run pytest -q
```

```
uv run applicant-ranking doctor
```

```
uv run applicant-ranking criteria lint
```

Commands: `criteria lint|draft`, `postings list`, `rank`, `evaluate`, `evaluate-file`, `backtest`, `fairness`, `export-audit`, `purge`, `hold`, `doctor`, `serve`. Run `uv run applicant-ranking --help` for the full list. Configuration uses `AR_*` environment variables; every one is documented in `.env.example`.

Safety model: the default mode is `dry_run`, which plans the write-back but never sends it. Both `dry_run` and `live` need approved criteria and an approved `config/policy.yaml` that passes for the posting's jurisdictions; `live` also needs `AR_WRITE_ENABLED=true`. The policy gate reads both the posting's location tags and Lever's structured address; a US posting with a location label it cannot map to a state or city is blocked until Lever has an address for it or Legal/HR list the label under `location_overrides`, and a New York location without a known non-NYC city counts as New York City. `evaluation` mode is for tuning and backtests: its output is watermarked and it can never write to Lever. Nothing can reject, archive, advance or hide a candidate.

Before first real run

1. **GCP:** pick an Egen project with Vertex AI enabled and set `AR_GCP_PROJECT`. Make sure application-default credentials belong to `nelma.perera@egen.ai`:

   ```
   gcloud auth login nelma.perera@egen.ai --update-adc
   ```

2. **Lever:** create a dedicated API key with only the permissions listed in docs/legal-hr-questions.md. Set `AR_LEVER_API_KEY` in `.env`, which is git-ignored.
3. **Criteria:** draft from a posting with `criteria draft --posting <id>`, have the hiring manager refine it, try it on sample resumes with `evaluate-file`, then have TA or HR approve it.
4. **Legal/HR:** copy `config/policy.example.yaml` to `config/policy.yaml` and get it completed and approved. Until then only `evaluation` mode runs.

Goal

Rank the applications to a Lever job posting against criteria written for the role, so recruiters can start with the strongest candidates. An LLM rates each resume against the criteria, code turns those ratings into a 1–100 score and an A–D tier, and the result goes back to Lever as tags and one advisory note. Recruiters make every decision. What the service must do is in docs/requirements.pdf; how it does it is in docs/design.md.

Decisions and open questions

Decided on 2026-10-08:

- **LLM provider:** Gemini Pro on Vertex AI (`gemini-3.1-pro-preview`, the only Pro model after `gemini-2.5-pro` retires on 2026-10-20). It is a Public Preview served only on the global endpoint, so there is no data-residency guarantee; Legal still needs to approve it.
- **Write-back:** tags that recruiters can filter on in Lever (the tier, and markers for applications that need a human check), plus one advisory note per application. Lever cannot sort by an integration-written score.
- **Measuring ranking quality:** `applicant-ranking backtest` compares the ranking with past recruiter decisions (AUC, precision@k, positive rate per tier) before any live use.
- **Where it runs:** a CLI for batch ranking, plus a webhook service for Cloud Run.

Still open (details in docs/legal-hr-questions.md):

- What will the per-role criteria contain, and which postings go first? The format is `criteria/*.yaml`; see the example file.
- Legal/HR: Illinois notice (in force), NYC LL144 scope, Colorado from 2027-01-01, the redaction and protected-term lists, retention, and approval of Vertex AI for resumes.

- Lever: a dedicated API key with the minimal permissions, whether it may read confidential postings, webhook signature tokens, and confirmation that the plan includes the Data API and Webhook features.

Environment

This folder is wired to the Egen identity: git commits as `nelma.perera@egen.ai` and GitHub as `nperera7` (via `~/.gitconfig-egen`). direnv pins gcloud to the `egen` configuration of the shared `~/.config/gcloud` store and gh to `~/.config/gh-egen`. GitHub pushes refuse to run until `nperera7` is logged into that gh store.